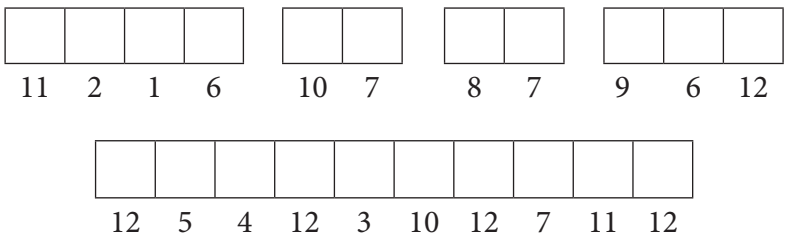
Name: _____

WORKSHEET

Inverse and determinant of a matrix

A failure is a person who has blundered, but is not able to ...



Elbert Hubbard

Rules

- Given $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$, the **determinant** of A, $\det A = ad - bc$.
- If $\det A = 0$, the matrix is **singular** and does not have an inverse.

Otherwise, the **inverse** of A, $A^{-1} = \dfrac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$

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Questions

Calculate the determinant of:

1 $\begin{bmatrix} 5 & -2 \\ -3 & 4 \end{bmatrix}$

2 $\begin{bmatrix} 7 & 11 \\ -2 & -3 \end{bmatrix}$

3 $\begin{bmatrix} 8 & -5 \\ -4 & 2 \end{bmatrix}$

4 $\begin{bmatrix} -12 & -4 \\ 9 & 3 \end{bmatrix}$

Calculate the inverse of:

5 $\begin{bmatrix} -3 & 2 \\ -7 & 4 \end{bmatrix}$

6 $\begin{pmatrix} -8 & 5 \\ 3 & \frac{1}{4} \end{pmatrix}$

7 $\begin{bmatrix} 2 & -3 \\ 9 & -11 \end{bmatrix}$

8 $\begin{bmatrix} 7 & -2 \\ 14 & -5 \end{bmatrix}$

9 $\begin{bmatrix} 2 & 10 \\ 1\frac{1}{8} & 6 \end{bmatrix}$

10 $\begin{bmatrix} -3 & 7 \\ 9 & -4 \end{bmatrix}$

11 $\begin{bmatrix} 15 & 6 \\ -10 & -4 \end{bmatrix}$

12 $\begin{bmatrix} -7 & 5 \\ -5 & 3\frac{1}{2} \end{bmatrix}$

Solutions

A 1

C Singular, no inverse

E $2\begin{bmatrix} 3\frac{1}{2} & -5 \\ 5 & -7 \end{bmatrix}$

H $-\frac{1}{17}\begin{bmatrix} \frac{1}{4} & -5 \\ -3 & -8 \end{bmatrix}$

I $-\frac{1}{51}\begin{bmatrix} -4 & -7 \\ -9 & -3 \end{bmatrix}$

N $\frac{1}{5}\begin{bmatrix} -11 & 3 \\ -9 & 2 \end{bmatrix}$

O $2\frac{1}{7}\begin{bmatrix} -5 & 2 \\ -14 & 7 \end{bmatrix}$

P 0

R -4

S 14

T $\frac{4}{3}\begin{bmatrix} 6 & 210 \\ 21\frac{1}{18} & 2 \end{bmatrix}$

X $\frac{1}{2}\begin{bmatrix} 4 & -2 \\ 7 & -3 \end{bmatrix}$

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